



NORTH SOUTH UNIVERSITY
DEPARTMENT OF ELECTRICAL & COMPUTER
ENGINEERING

Assignment-2

Summer 2025

EEE141: Electrical Circuits I
Section:

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Assignment-2

Attach your cover page and also this Question paper

Book: Introductory Circuit Analysis by Boylestad 11th Edition

Chapter 6 Exercise Problems (11th Edition, check the PDF):

14,15,16,18,21,22,24,25,26,27,28,29,30,38,39,40

[10 Marks for Problems 28 & 30; for rest of the Problems 4 Marks]

Total 16 Math Problems. Marks Distribution: $(14 \times 4 = 56) + (2 \times 10 = 20) = 76$ Marks

Additional 4 marks for overall format.

Total marks $76 + 4 = 80$ marks

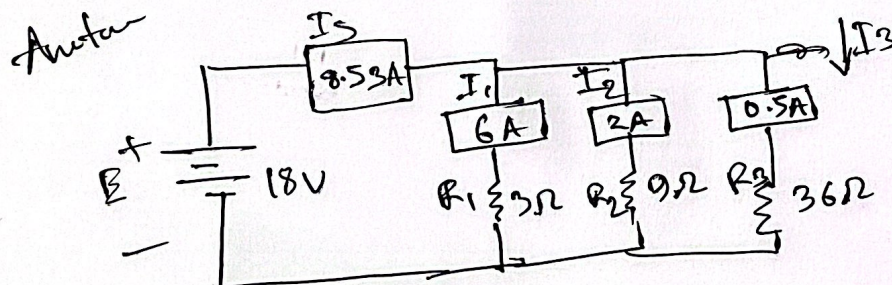
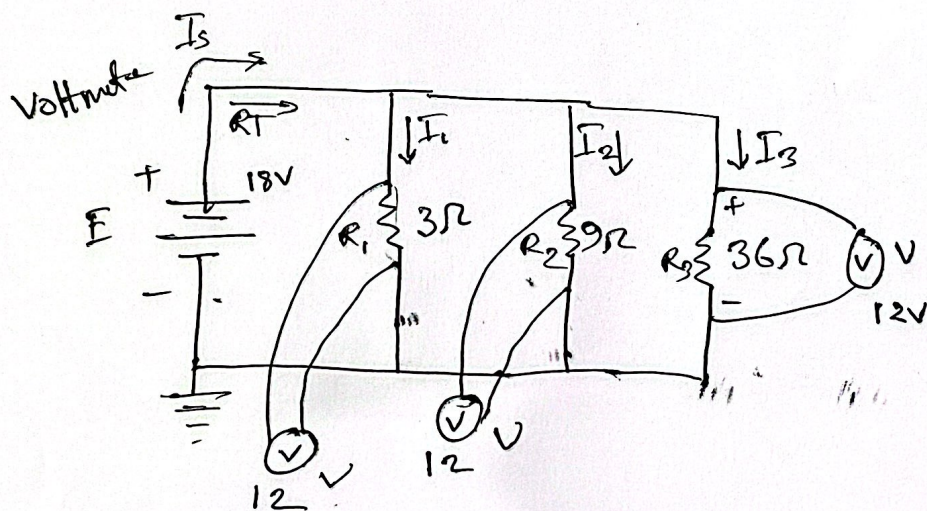
Add Cover Page, Question Paper, Write down all the questions and Redraw all the circuits with proper voltage polarity and current direction.

Submission offline/class time: 6th July 2025

14.

a) Redraw the network and insert ammeters to measure the source current and the current through each branch.

b) Connect a voltmeter to measure the source voltage and the voltage across resistor, R_3 . Is there any difference in the connections? why?



$$I_s = I_1 + I_2 + I_3$$

$$= 8.5A$$

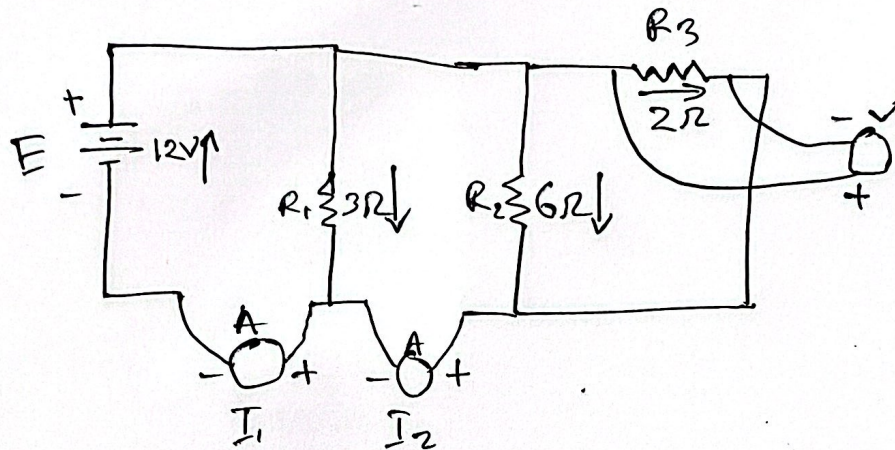
$$I_1 = \frac{18}{3} = 6A$$

$$I_2 = \frac{18}{9} = 2A$$

$$I_3 = \frac{18}{36} = 0.5A$$

Ans

15. what is the response of the voltmeter and ammeters connected.



here R_1 and R_2 is connected in parallel

$$\therefore R_A = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{18}{9} = 2\Omega$$

R_A and R_3 is parallel

$$\therefore R_T = \frac{R_3 \times R_A}{R_3 + R_A} = \frac{4}{4} = 1\Omega$$

$$I_s = \frac{E}{R_T} = \frac{12}{1} = 12A$$

$$I_{R_1} = \frac{12}{3} = 4A$$

$$I_{R_2} = \frac{12}{6} = 2A$$

$$I_{R_3} = \frac{12}{2} = 6A$$

Now, For the Ammeter 2, current is the sum of R_2 and R_3

$$\begin{aligned} I_2 &= IR_2 + IR_3 \\ &= 2 + 6 = 8A \end{aligned}$$

$$\begin{aligned} \text{For } I_1 &= IR_1 + IR_2 + IR_3 \\ &= 4 + 2 + 6 \\ &= 12A \end{aligned}$$

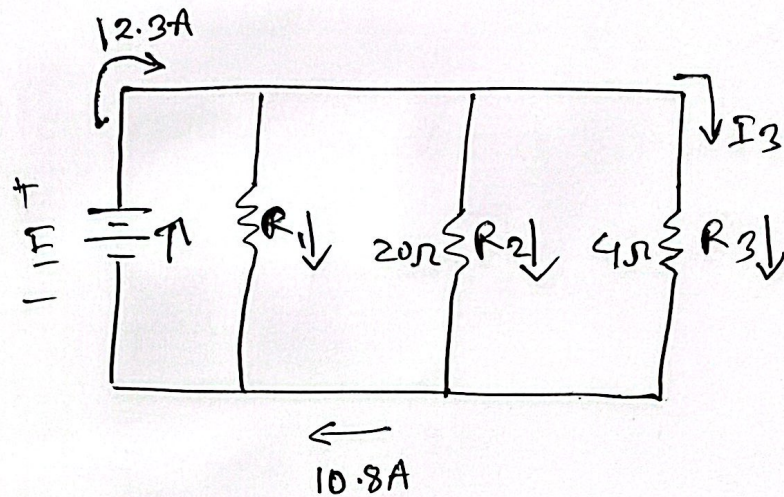
in parallel circuit voltage is same for each resistance.

So, the voltage of the R_3 resistor is 12V

Ans.

16. Given the information provided in Fig.

Find the unknown quantities: E , R_1 , and I_3 .



$$R_2 \text{ and } R_3 = \frac{R_2 \times R_3}{R_2 + R_3} = 3.33\Omega$$

voltage drop here $I_2 \times 3.33\Omega = 36V$

voltage in the circuit is $36V$. $R_T = \frac{36}{12.3} = 2.93\Omega$

current through $I_3 = \frac{V}{R_3} = \frac{36}{4} = 9A$

For R_1 , $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$R_1 = \frac{1}{\frac{1}{R_T} - \frac{1}{R_2} - \frac{1}{R_3}} = \frac{1}{\frac{1}{2.93} - \frac{1}{20} - \frac{1}{4}}$$

$$R_1 = 24\Omega$$

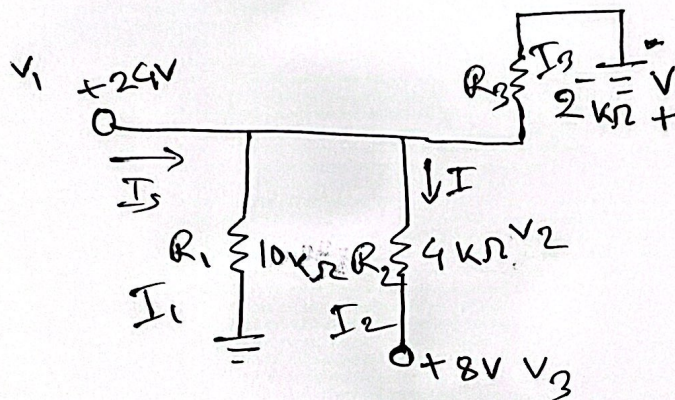
Am.

18. For the network.

a) Find the current I .

b) Determine the voltage v .

c) calculate the source current I_s .



a) voltage in $R_2 = (V_1 - V_3) = 16V$

current through here $I_2 = \frac{16}{4k\Omega} = 4mA$

b) In parallel circuit voltage is same so the voltage v is 24V

c) current $I_{R_1} = \frac{24}{10k\Omega} = 2.4mA$

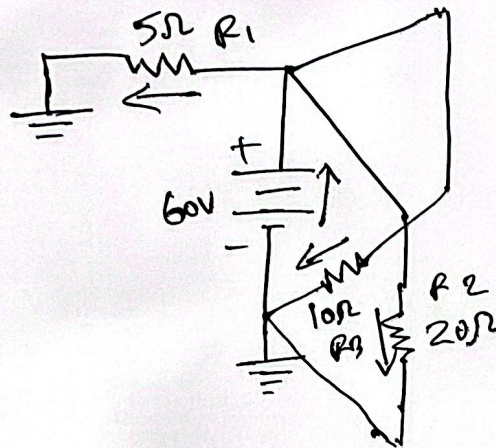
$$I_{R_2} = \frac{24}{2k\Omega} = 12mA$$

source current $I_s = I_1 + I_2 + I_3$

$$= 18.4mA$$

Ans.

21. Determine the power delivered by the dc battery.



$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$R_T = \frac{1}{7/20}$$

$$R_T = 2.86 \Omega$$

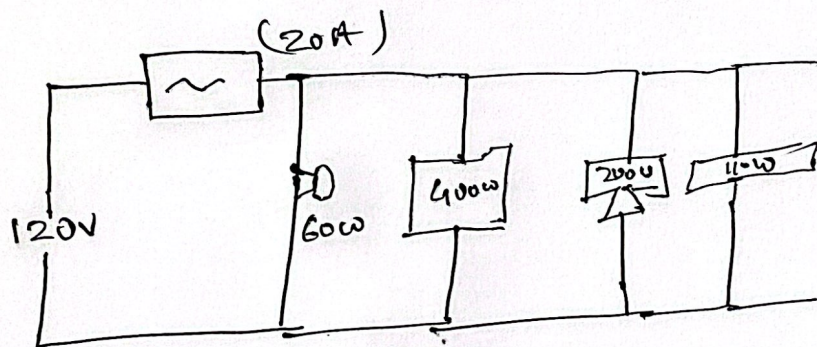
$$I_s = \frac{E}{R_T} = \frac{60}{2.86 \Omega} = 20.98 \text{ A}$$

$$\text{power } P = E \times I_s = 60 \times 20.98 = 1.26 \text{ kW}$$

Ans.

22.

- a) Determine the current through each parallel branch of the system.
- b) Calculate the current drawn from the 120V source. Will the 20A breaker trip?
- c) What is the total resistance of the network?
- d) Determine the power delivered by the source.



$$a) \text{ bulb } I = \frac{P_b}{E} = \frac{600}{120} = 5 \text{ A}$$

$$\text{washer } I = \frac{400}{120} = 3.33 \text{ A}$$

$$\text{TV } I = \frac{200}{120} = 1.67 \text{ A}$$

$$\text{DVD } I = \frac{110}{120} = 0.92 \text{ A}$$

$$b) \text{ Total } I = I_1 + I_2 + I_3 + I_4$$

$$= 10.92 \text{ A}$$

$$c) R_T = \frac{E}{I_s} = \frac{120}{10.92} = 10.989 \Omega$$

$$d) P_s = E \times I_s$$

$$= 120 \text{ V} \times 10.92 \text{ A}$$

$$= 1.31 \text{ kW}$$

No ω

$$P_L = P_1 + P_2 + P_3 + P_4$$

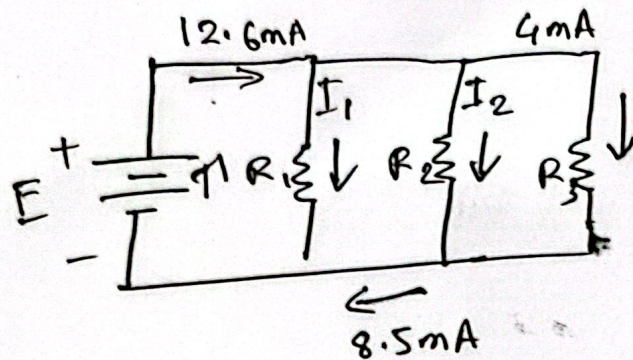
$$= (600 + 400 + 200 + 110)$$

$$= 1310 \text{ W}$$

$$= 1.31 \text{ kW}$$

Am.

24. Using KCL, determine the unknown currents for the parallel network.



$$I = I_1 + I_2 + I_3$$

$$12.6 = I_1 + 8.5$$

$$I_1 = (12.6 - 8.5) \text{ mA}$$

$$= 4.1 \text{ mA}$$

$$\text{Again } I = I_1 + I_2 + I_3$$

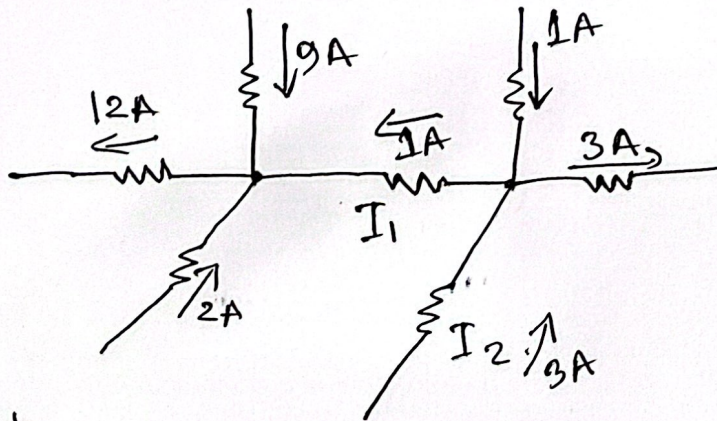
$$12.6 = 4.1 + I_2 + 4$$

$$I_2 = 12.6 - 8.1$$

$$= 4.5 \text{ mA}$$

Ans.

25. using KCL find the unknown current.



1st

$$I_1 + 9A + 2 = 12$$

$$I_1 = 1A$$

hence I_1 Entering.

2nd,

$$1 + I_2 = I_1 + 3$$

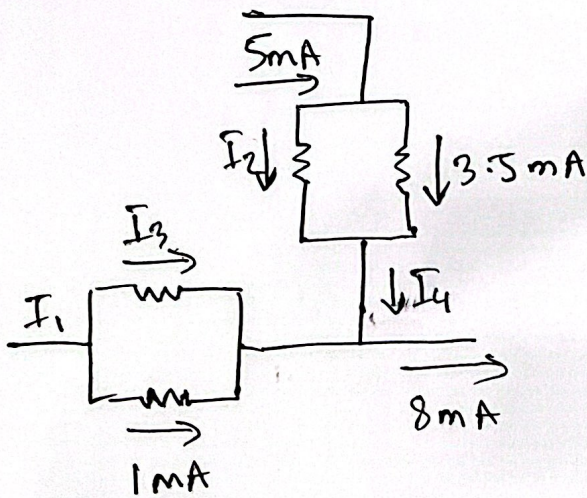
$$I_2 = 4 - 1$$

$$I_2 = 3A$$

hence I_2 Entering

Ans.

26. Using KCL determine the unknown current.



$$5 = 3.5 + I_2$$

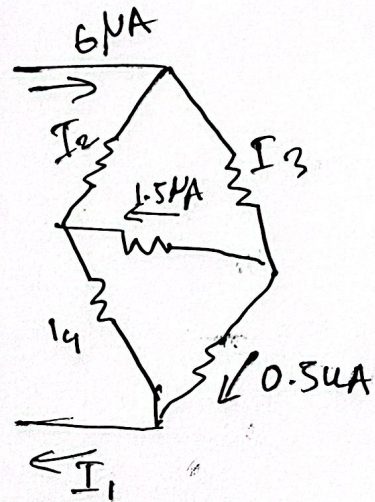
$$I_2 = 5 - 3.5 = 1.5 \text{ mA}$$

$$I_4 = 5 \text{ mA}$$

$$I_3 + I_4 + I_5 = 8 - 5 = 1 + I_3$$

$$I_3 = 2$$

$$I_1 = 3$$



$$I_3 = 1.5 + 0.5 \mu A$$
$$= 2 \mu A$$

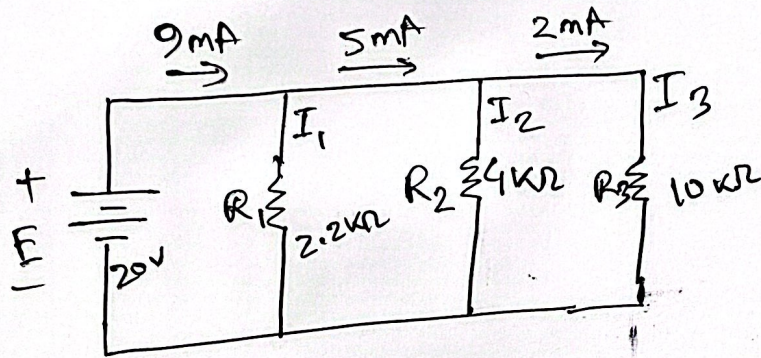
$$I_2 = 6 - 2 = 4 \text{ mA}$$

$$I_4 = I_2 + 1.5$$
$$= 4 + 1.5 = 5.5 \text{ mA}$$

$$I_1 = 14 + 0.5$$
$$= 6 \mu A$$

Am.

27. Using the information, Find the R_1 , R_2 and R_T . voltage E .



voltage in R_2 resistor $V = I_2 \times R$
 $= 5 \times 4 = 20V$

in parallel circuit voltage is same

$$R_1 = \frac{V}{I_1} = \frac{20}{9} = 2.22k\Omega$$

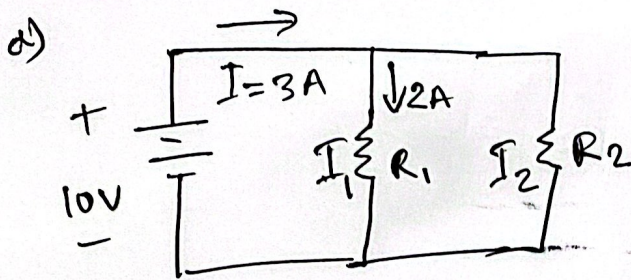
$$R_3 = \frac{V}{I_3} = \frac{20}{2} = 10k\Omega$$

$$E = 20V$$

Aw.

练.

28. Find the unknown quantities



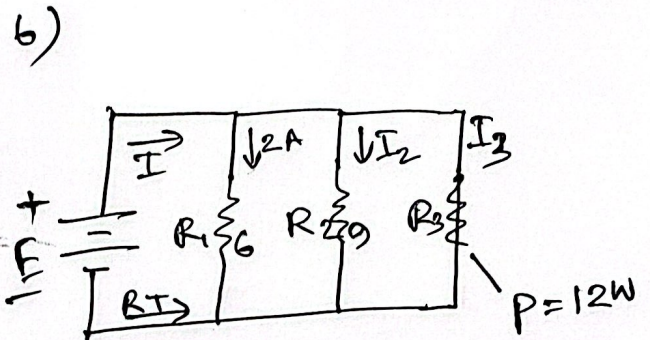
Kcl, $I = I_1 + I_2$

$I_2 = I_A$

$E = V_1 = V_2 = 10V$

$R_1 = \frac{10}{2} = 5\Omega$

$R_2 = \frac{10}{1} = 10\Omega$



$V_1 = I_1 \times R_1$

$= 2 \times 6 = 12V$

parallel $E = V_1 = V_2 = V_3$

$I_2 = \frac{V_2}{R_2} = \frac{12}{9} = 1.33A$

Current $I_3 = \frac{P_3}{V_3} = 1A$

$\therefore R_3 = \frac{V_3}{I_3} = 12\Omega$

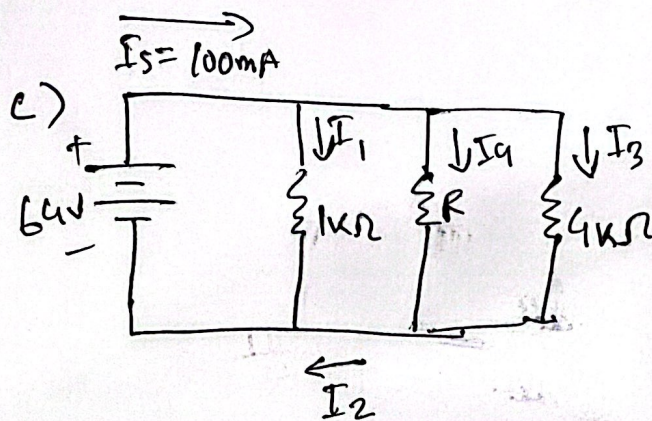
$I_s = 4.33 A$

$R_T = \frac{E}{I} = \frac{12}{4.33}$

$= 2.78\Omega$

Ans.

28.



$$I_1 = \frac{V_1}{R_1} = \frac{64V}{1k\Omega}$$

$$= 64mA$$

$$I_3 = \frac{V}{R_3} = \frac{64}{4}$$

$$= 16mA$$

$$I_s = I_1 + I_4 + I_3$$

$$I_4 = 20mA$$

$$R = \frac{V}{I_4} = \frac{64}{20mA}$$

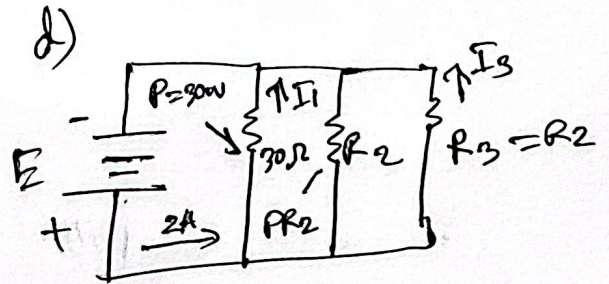
$$= 3.2k\Omega$$

$$I_2 = I_4 + I_3$$

$$= 20 + 16$$

$$= 36mA$$

Ans.



$$P = V_1 I_1$$

$$= V_1 \times \frac{V_1}{R_1}$$

$$V_1 = \sqrt{P_1 \times R_1}$$

$$= \sqrt{30W \times 30\Omega}$$

$$= 30V$$

$$I_1 = \frac{V}{R_1} = \frac{30}{30} = 1A$$

$$I = I_1 + 2I_4$$

$$I_4 = 0.5 = I_2 = I_3$$

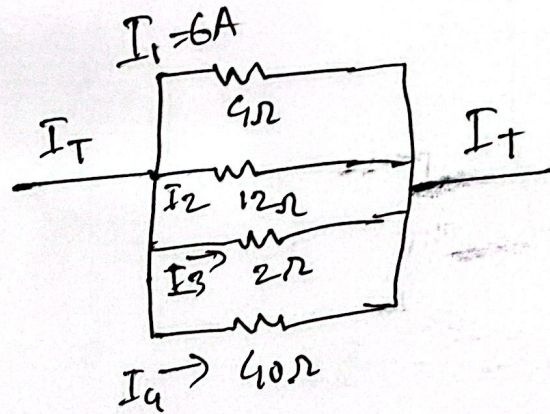
$$R_2 = R_3 = \frac{30}{0.5} = 60\Omega$$

$$P_{R_2} = V_2 \times I_2$$

$$= 30 \times 0.5A$$

$$= 15W$$

29. Determine all the currents. Do not use Ohm's law



$$I_n = I_T \frac{R_T}{R_n + R_{all}}$$

$$\frac{I_1}{1/R_1} = \frac{I_2}{1/R_2} = \frac{I_3}{1/R_3}$$

$$= 2 : 1 : 6$$

$$I_1 : I_2 : I_3 = 2 : 1 : 6$$

$$I = 6A$$

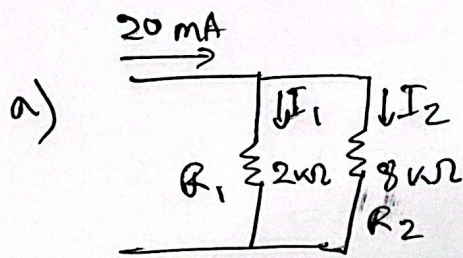
$$I_2 = 1 \times 3 = 3A$$

$$I_3 = 6 \times 3 = 18A$$

$$I_4 = 0.6A$$

Ans.

30. Determine the current.



$$I_1 = \left(\frac{R_2}{R_1 + R_2} \right) \times I_T$$

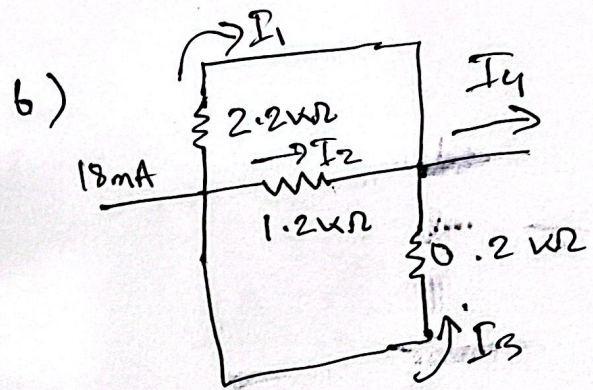
$$= \frac{8}{10} \times 20$$

$$= 16 \text{ mA}$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2} \right) \times I_T$$

$$= \frac{2}{10} \times 20$$

$$= 4 \text{ mA}$$



$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$= 159.01 \Omega$$

$$I_1 = \frac{I_T \times R_T}{R_1}$$

$$= 1.30 \text{ mA}$$

$$I_2 = 2.30 \text{ mA}$$

$$I_3 = 14.31 \text{ mA}$$

using KCL

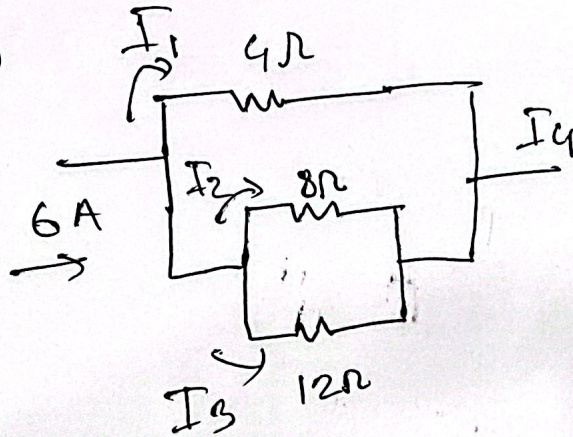
$$I_4 = I_1 + I_2 + I_3$$

$$= 18 \text{ mA}$$

Ans.

30.

c)



$$R_T = 2.182 \Omega$$

using CDR

$$I_1 = \frac{I_T \times R_T}{R_1} = \frac{6 \times 2.18}{4} = 3.27 \text{ A}$$

$$I_2 = \frac{I_T \times R_T}{R_2} = 1.64 \text{ A}$$

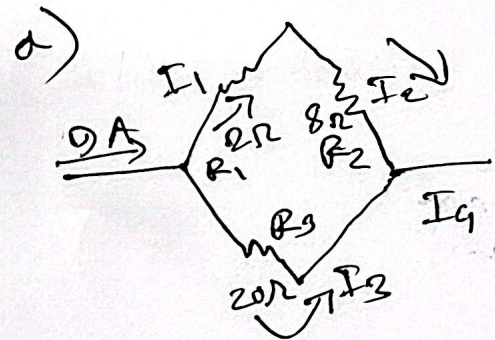
$$I_3 = 1.09 \text{ A}$$

vel

$$I_4 = I_1 + I_2 + I_3$$

$$= 6 \text{ A}$$

A



here R_1 and R_2 is series

$$\text{so } R_T = 8 + 2 \parallel 20$$

$$= 6.67$$

$$V = 9 \times 6.67$$

$$= 60 \text{ V}$$

$$I_3 = \frac{60}{20} = 3 \text{ A}$$

$$I_2 = I_1 = I_4 = \frac{60}{10} = 6 \text{ A}$$

$$I = 3 + 6$$

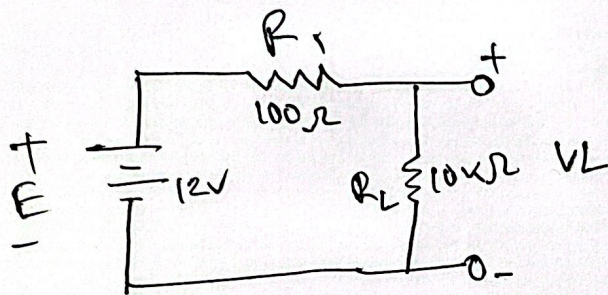
$$= 9 \text{ A}$$

Ans

38. a) Determine I_S and V_L .

b) " I_S if R_L is shorted.

c) " V_L if R_L is Replaced by open circuit.



$$a) I_S = \frac{E}{R_T} = \frac{12}{10 + 0.1} \\ = 1.188 \text{ mA}$$

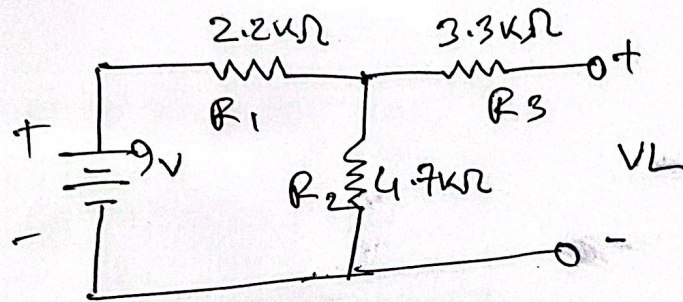
$$\text{Using VDR } V_L = \frac{R_L}{R_L + R_1} \times E \\ = \frac{10}{10.1} \times 12 \\ = 11.88 \text{ V}$$

$$\text{Ans. b) } I_S = \frac{E}{R_1} \\ = 0.12 \text{ A}$$

c) if R_L is ~~sh~~ open current can't flow. the voltage will remain same $V_L = E = 12 \text{ V}$.

Ans.

- 39.
- Determine the open circuit voltage V_L .
 - If $2.2\text{ k}\Omega$ is short. what is the new V_L
 - Determine V_L , if the $4.7\text{ k}\Omega$ is open circuit.



$$\begin{aligned}
 a) \quad V_L &= \frac{R_2}{R_1 + R_2} \times E \\
 &= \frac{4.7}{4.7 + 2.2} \times 9 \\
 &= 6.13\text{ V}
 \end{aligned}$$

b) If R_1 short E parallel to R_2 , so $V_L = 9\text{ V}$

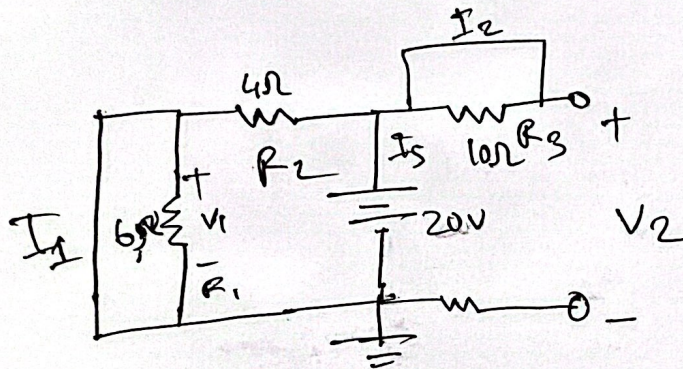
c) if R_2 is open circuit. it remains the same voltage. the voltage will be

$$V_L = 9\text{ V}$$

Ans.

40.

- The short-circuit currents I_1 and I_2
- The voltage V_1 and V_2
- The source current I_S



$$a) I_1 = \frac{E}{R_2} = \frac{20}{4} = 5A$$

I_2 is open circuit, so the current is 0

$$I_2 = 0A$$

- ~~I_1~~ $I_1 \parallel R_1$, so current direct flow via I_1

$$\text{So, } V_1 = 0V$$

V_2 is open circuit it remains same with E

$$\text{So, } V_2 = 20V$$

- $I_S = I_1 = 5A$

Ans.